Graduate Analysis 1: midsem exam

Aditya Padiyar

October 2026

Definition 1. *A sequence of functions f_n converges almost uniformly to f if for every $\epsilon > 0$ there is a set A_ϵ with $\mu(A_\epsilon) < \epsilon$ such that f_n converges uniformly to f on $(A_\epsilon)^c$*

Definition 2. *f is integrable if $\int |f| < \infty$*

Definition 3. *A sequence of integrable functions f_n is **uniformly integrable** if for all $\epsilon > 0$ there is $\delta >$ such that $\int_A |f_n| < \epsilon$ for all sets A with $\mu(A) < \delta$*

Claim 1. *On a finite measure space, convergence almost everywhere is equivalent to convergence almost uniformly*

Claim 2. *Suppose $f_n^0, f_n^1, f_n^2, f_0, f_1, f_2$ are functions such that*

- *f_n^i, f_0, f_2 are integrable*
- *$f_n^0 \leq f_n^1 \leq f_n^2$*
- *$f_n^i \to f_i$*
- *$\int f_n^0 \to \int f_0$ and $\int f_n^2 \to \int f_2$*

Then we have

- *f_1 is integrable*
- *$\int f_n^1 \to \int f_1$*

Claim 3. *Suppose $f : \mathbb{R}^2 \to \mathbb{R}$ is measurable and satisfies:*

- *$x \to f(x_0, x)$ is integrable for all x_0*
- *$g(x, y) = \partial_x f(x, y)$ exists everywhere*
- *There is an integrable real valued function h on $\mathbb{R}$ such that $|g(x, y)| \leq h(y)$*

Let $\phi(x) = \int f(x, y)dy$. Then we have that

- ϕ *is differentiable*
- $\phi'(x) = \int g(x, y)dy$

Claim 4. *If f_n are uniformly integrable and converge pointwise to f, then we have:*

- f *is integrable*
- $\int f_n \to \int f$

Claim 5. *Suppose X is a random variable with finite expectation and $F \subseteq \mathbb{R}$ is closed. Suppose that whenever E has positive probability $\frac{\mathbb{E}[X1_E]}{P(E)} \in F$. Then $P(X \in F) = 1$*